

IT 4090

Cloud Computing

QnA

01. You have successfully set up a VPC peering connection in your account between two VPCs-VPC A and VPC B, each in a different region. When you are trying to make a request from VPC A to VPC B, request getting failed. Which of the following could be the reason?

- A. Cross region peering is not supported in AWS.
- B. CIDR blocks of both VPCs might be overlapping.
- C. Routes not configured in route tables for peering connections.
- D. VPC A security group default outbound rules not allowing traffic to VPC B IP range.

Answer: C

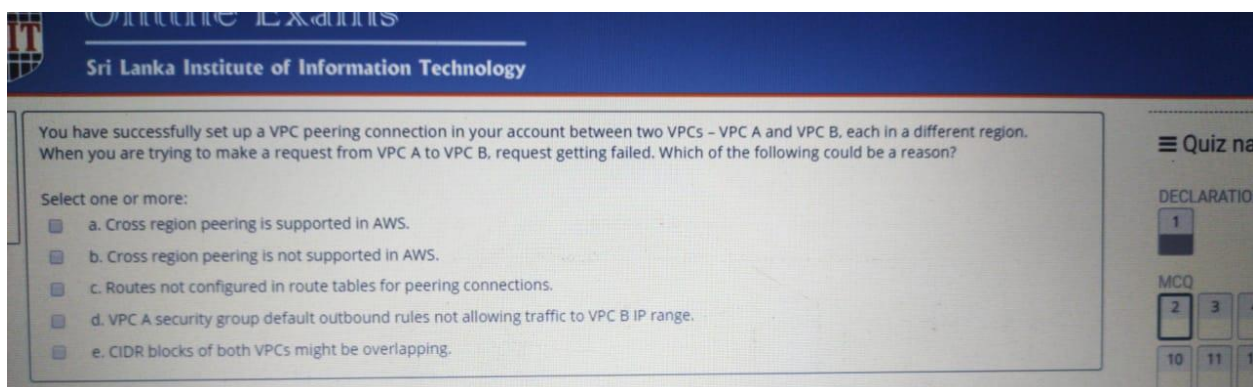
**Option A is not correct.** Cross region VPC peering is supported in AWS.

**Option B is not correct.**

When the VPC IP CIDR blocks are overlapping, you cannot create a peering connection. Question states the peering connection was successful.

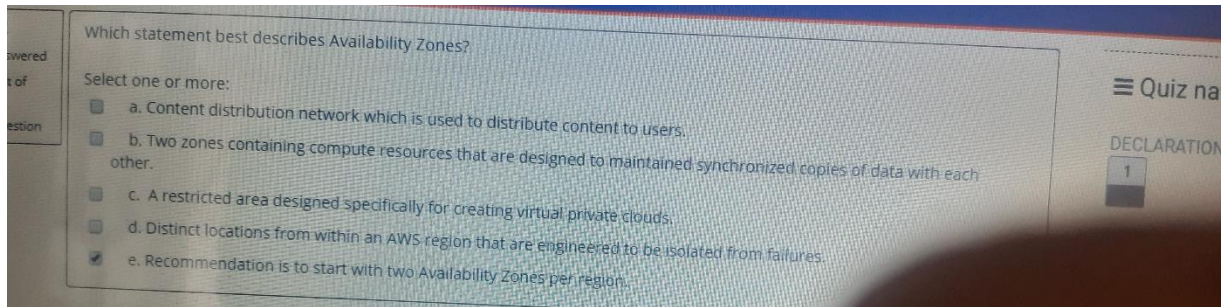
**Option C is correct.** To send private IPv4 traffic from your instance to an instance in a peer VPC, you must add a route to the route table that's associated with your subnet in which your instance resides. The route points to the CIDR block (or portion of the CIDR block) of the peer VPC in the VPC peering connection.

**Option D is not correct.** A security group's default outbound rule allows all traffic going out from the resources attached to the security group.



02. Which statement describes Availability Zones?

- D. Distinct locations from within an AWS region that are engineered to be isolated from failures.

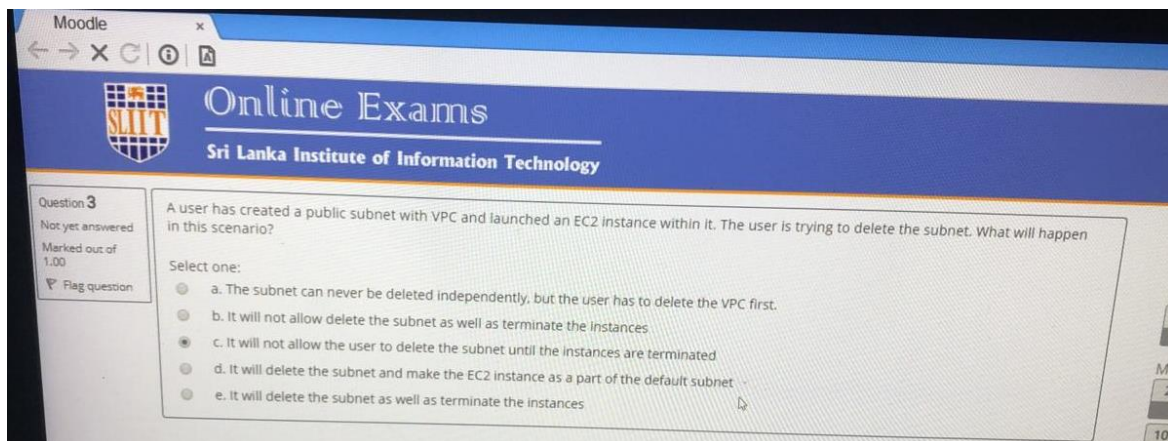


03. A user has created a public subnet with VPC and launched an EC2 instance within it. The user is trying to delete the subnet. What will happen in this scenario?

- A. It will delete the subnet and make the EC2 instance as a part of the default subnet
- B. It will not allow the user to delete the subnet until the instances are terminated**
- C. It will delete the subnet as well as terminate the instances
- D. The subnet can never be deleted independently, but the user has to delete the VPC first

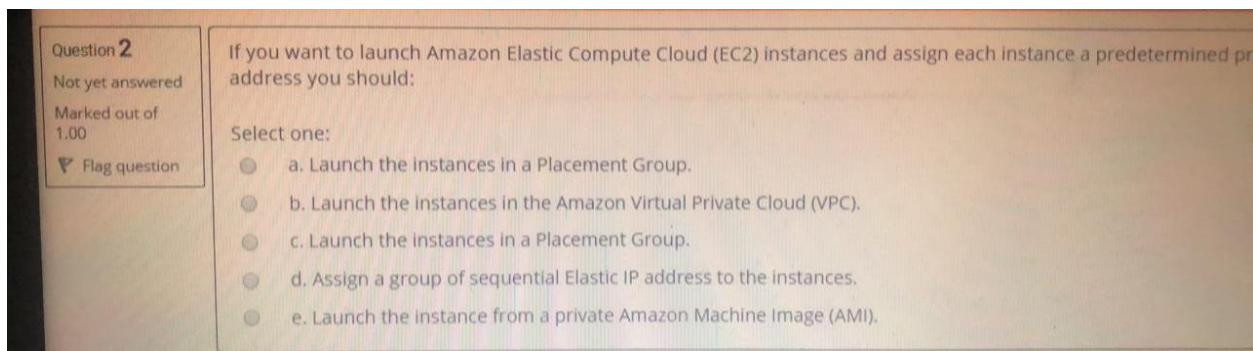
Explanation:

A Virtual Private Cloud (VPC) is a virtual network dedicated to the user's AWS account. A user can create a subnet with VPC and launch instances inside that subnet. When an instance is launched it will have a network interface attached with it. The user cannot delete the subnet until he terminates the instance and deletes the network interface.



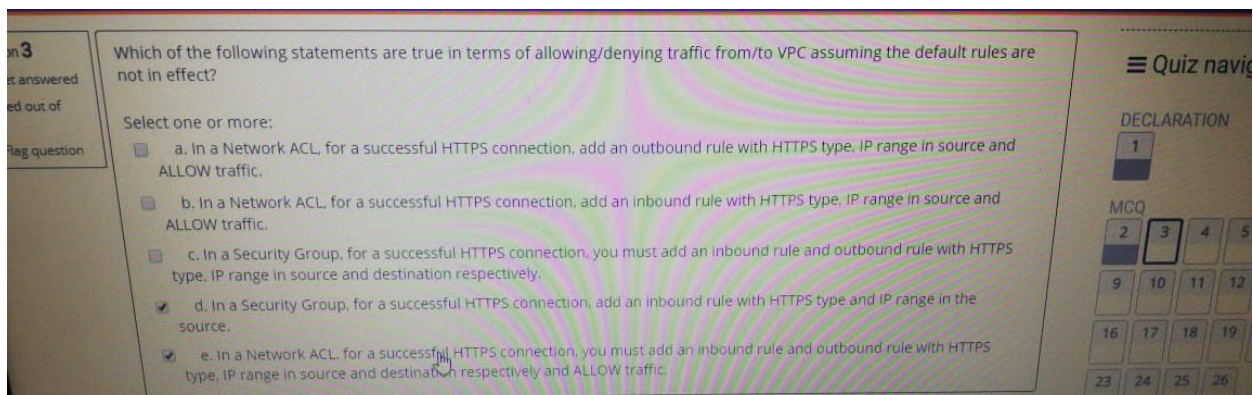
04. If you want to launch Amazon Elastic Compute Cloud (EC2) instances and assign each instance a predetermined private IP address you should:

- A. Launch the instance from a private Amazon Machine Image (AMI).
- B. Assign a group of sequential Elastic IP address to the instances.
- C. Launch the instances in the Amazon Virtual Private Cloud (VPC).
- D. Launch the instances in a Placement Group.
- E. Use standard EC2 instances since each instance gets a private Domain Name Service (DNS) already.



05. Which of the following statements are true in terms of allowing/denying traffic from/to VPC assuming the default rules are not in effect? (choose multiple)

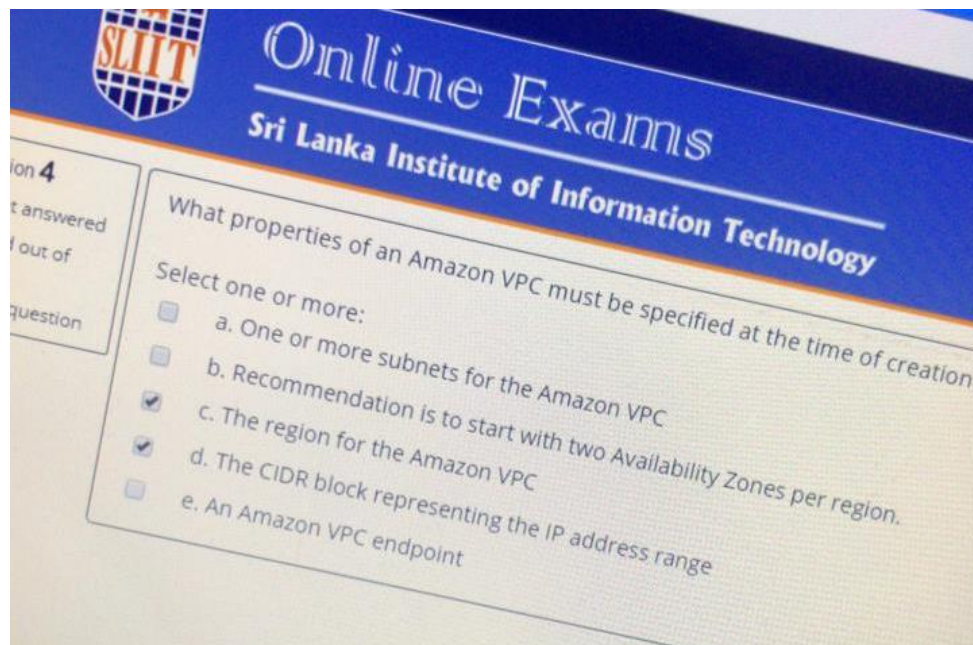
- A. In a Network ACL, for a successful HTTPS connection, add an inbound rule with HTTPS type, IP range in source and ALLOW traffic.
- B. In a Network ACL, for a successful HTTPS connection, you must add an inbound rule and outbound rule with HTTPS type, IP range in source and destination respectively and ALLOW traffic
- C. In a Security Group, for a successful HTTPS connection, add an inbound rule with HTTPS type and IP range in the source.
- D. In a Security Group, for a successful HTTPS connection, you must add an inbound rule and outbound rule with HTTPS type, IP range in source and destination respectively.



06. What properties of an Amazon VPC must be specified at the time of creation?

- A. The CIDR block representing the IP address range
- B. One or more subnets for the Amazon VPC
- C. The region for the Amazon VPC
- D. Amazon VPC Peering relationships
- E. Recommendation is to start with two Availability Zones per region

*A, C. The CIDR block is specified upon creation and cannot be changed. An Amazon VPC is associated with exactly one region which must be specified upon creation. You can add a subnet to an Amazon VPC any time after it has been created, provided its address range falls within the Amazon VPC CIDR block and does not overlap with the address range of any existing CIDR block. You can set up peering relationships between Amazon VPCs after they have been created.*



07. When attached to an Amazon VPC, which two components provide connectivity with external networks?

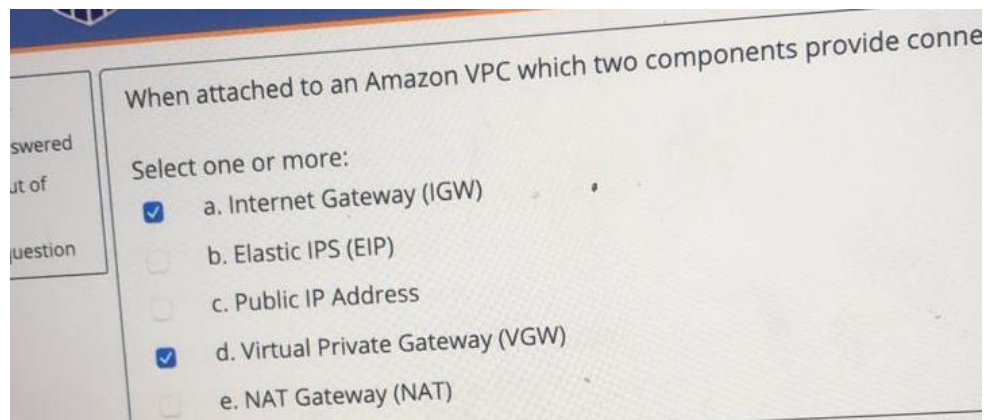
- A. Elastic IPS (EIP)

B. NAT Gateway (NAT)

C. Internet Gateway (IGW)

D. Virtual Private Gateway (VGW)

E. Public IP Address



08. A company has configured and peered two VPCs: VPC-1 and VPC-2. VPC-1 contains only private subnets and VPC-2 contains only public subnets. The company uses a single AWS Direct Connect Connection and private virtual interface to connect their on-premises network with VPC-1. What are the methods that increase the fault tolerance of the connection to VPC-1?

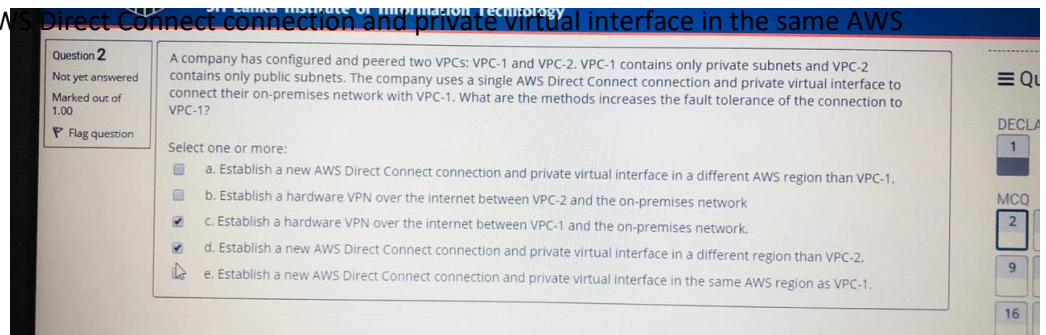
A. Establish a hardware VPN over the internet between VPC-2 and the on-premises network.

B. Establish a hardware VPN over the internet between VPC-1 and the on-premises network.

C. Establish a new AWS Direct Connect connection and private virtual interface in the same region as VPC-2.

D. Establish a new AWS Direct Connect connection and private virtual interface in a different AWS region than VPC-1.

E. Establish a new AWS Direct Connect connection and private virtual interface in the same AWS region as VPC-1.



09. What are the components of VPC?

A. Subnets

B. VPN

- C. NAT Gateway
- D. Internet Gateway
- E. Direct Connect

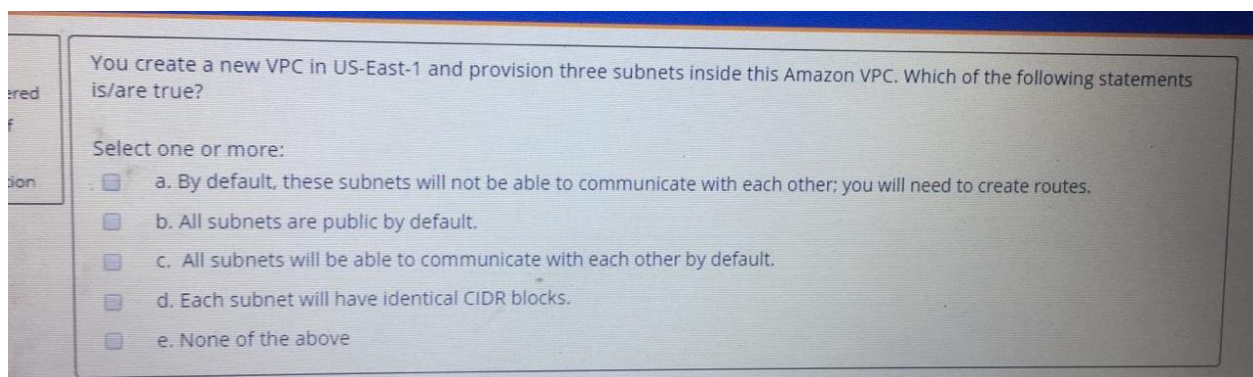
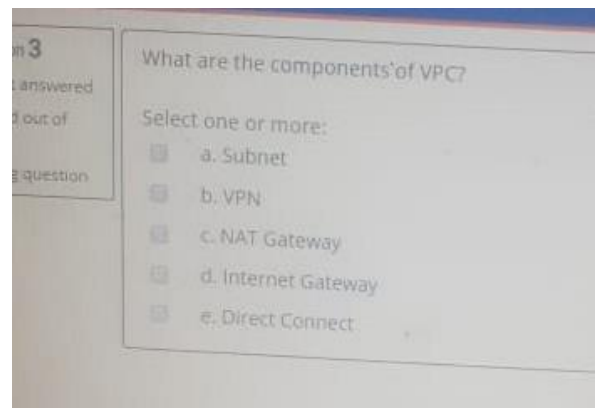
#### 5 Main Components of a VPC

- Internet Gateways (or Virtual Private Gateways)
- Route Tables
- NACLs
- Subnets
- Security Groups

10. You create a new VPC in US-East-1 and provision three subnets inside this Amazon VPC. Which of the following statements is/are true?

- A. By default, these subnets will not be able to communicate with each other; you will need to create routes.
- B. All subnets are public by default.
- C. All subnets will be able to communicate with each other by default.
- D. Each subnet will have identical CIDR blocks.

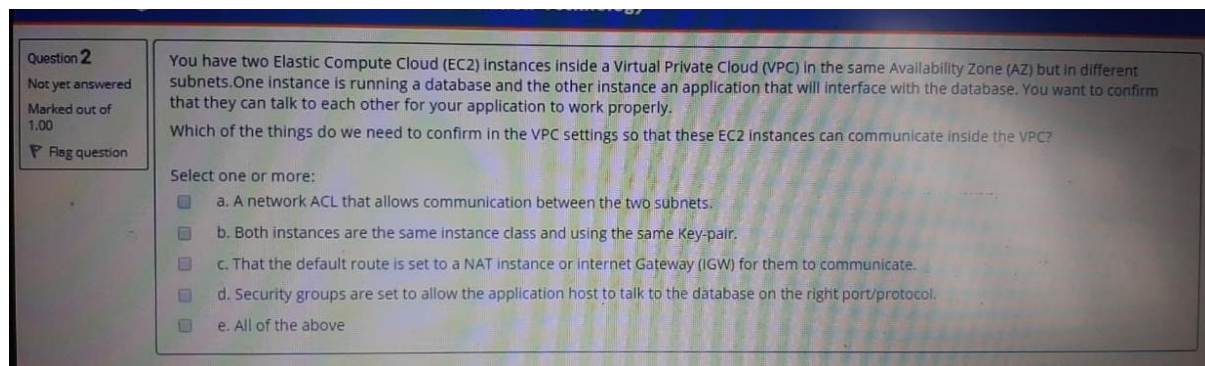
Ans : C. When you provision an Amazon VPC, all subnets can communicate with each other by default.



11. You have two Elastic Compute Cloud (EC2) instances inside a Virtual Private Cloud (VPC) in the same Availability Zone (AZ) but in different subnets. One instance is running a database and the other instance an application that will interface with the database. You want to confirm that they can talk to each other for your application to work properly.

Which two things do we need to confirm in the VPC settings so that these EC2 instances can communicate inside the VPC? (Choose two.)

- A. A network ACL that allows communication between the two subnets.
- B. Both instances are the same instance class and using the same Key-pair.
- C. That the default route is set to a NAT instance or internet Gateway (IGW) for them to communicate.
- D. Security groups are set to allow the application host to talk to the database on the right port/protocol



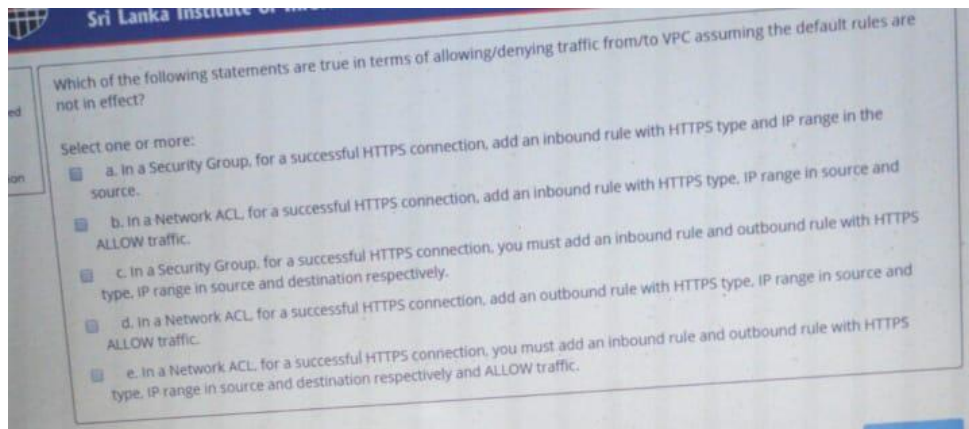
12. Which of the following statements are true in terms of allowing/denying traffic from/to VPC assuming the default rules are not in effect?

- A. In a Network ACL, for a successful HTTPS connection, add an inbound rule with HTTPS type, IP range in source and ALLOW traffic.
- B. In a Network ACL, for a successful HTTPS connection, you must add an inbound rule and outbound rule with HTTPS type, IP range in source and destination respectively and ALLOW traffic.
- C. In a Security Group, for a successful HTTPS connection, add an inbound rule with HTTPS type and IP range in the source.
- D. In a Security Group, for a successful HTTPS connection, you must add an inbound rule and

outbound rule with HTTPS type, IP range in source and destination respectively.

Answer: B, C

- Security groups are stateful — if you send a request from your instance, the response traffic for that request is allowed to flow in regardless of inbound security group rules.
- Responses to allowed inbound traffic are allowed to flow out, regardless of outbound rules.
- Network ACLs are stateless; responses to allowed inbound traffic are subject to the rules for outbound traffic (and vice versa).
- Option A is not correct. NACL must have an outbound rule defined for a successful connection due to its stateless nature.
- Option B is correct.
- Option C is correct.
- Configuring an inbound rule in security group is enough for a successful connection due to its stateful nature.
- Option D is not correct.
- Configuring an outbound rule for incoming connection is not required in security groups



### 13. What happens when you create a new Amazon VPC?

A. A main route table is created by default.

B. Three subnets are created by default—one for each Availability Zone.

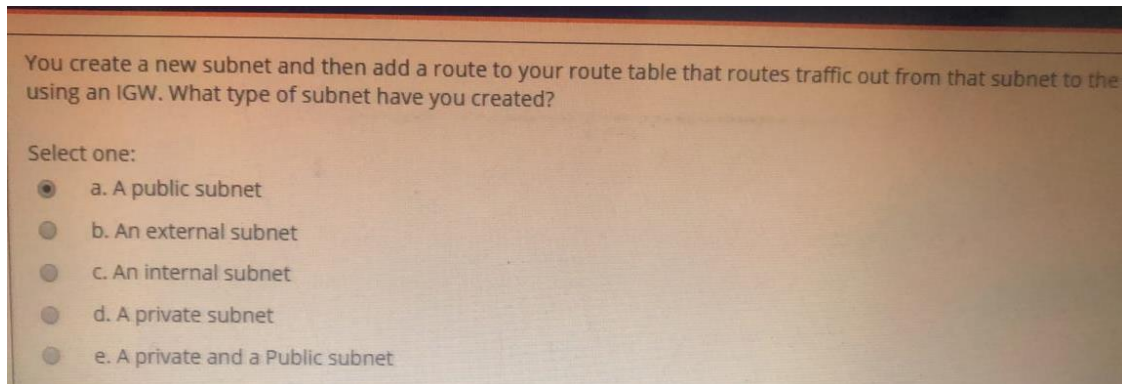
C. Three subnets are created by default in one Availability Zone.

D. An IGW is created by default.

A - When you create an Amazon VPC, a route table is created by default. You must manually create subnets and an IGW.

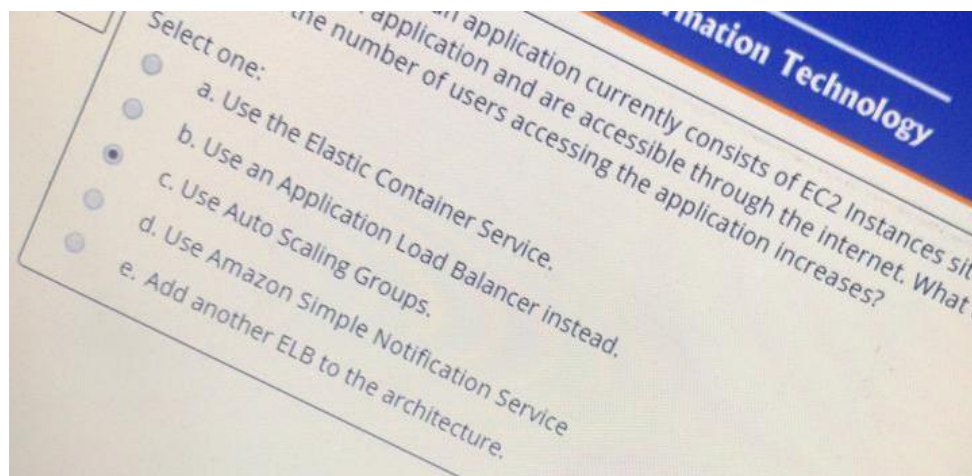
14. You create a new subnet and then add a route to your route table that routes traffic out from that subnet to the Internet using an IGW. What type of subnet have you created?

- A. An internal subnet
- B. A private subnet
- C. An external subnet
- D. A public subnet



15. Your architecture for an application currently consists of EC2 Instances sitting behind a classic ELB. The EC2 Instances are used to serve an application and are accessible through the internet. What can be done to improve this architecture in the event that the number of users accessing the application increases?

- A. Add another ELB to the architecture.
- B. Use Auto Scaling Groups.
- C. Use an Application Load Balancer instead.
- D. Use the Elastic Container Service
- AWS Documentation mentions the following:
  - AWS Auto Scaling monitors your applications and automatically adjusts capacity to maintain steady, predictable performance at the lowest possible cost. Using AWS Auto Scaling, it is easy to setup application scaling for multiple resources across multiple services in minutes.

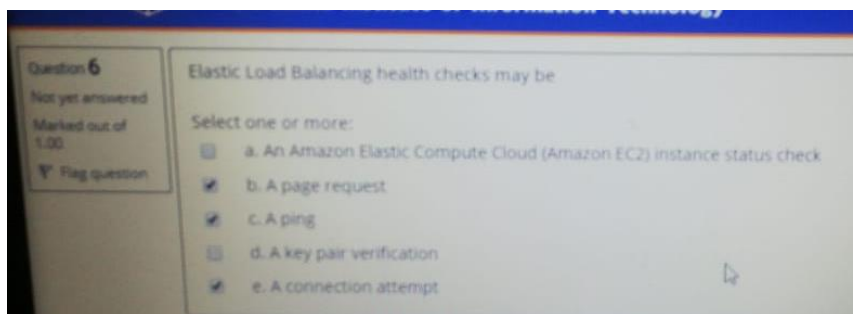


16. Which of the following are required elements of an Auto Scaling group? (Choose 2 answers)

- A. Minimum size
- B. Health checks
- C. Desired capacity
- D. Launch configuration

An Auto Scaling group must have a minimum size and a launch configuration defined in order to be created. Health checks and a desired capacity are optional.

17. Elastic Load Balancing health check may be



18. A user has configured the Auto Scaling group with the minimum capacity as 3 and the maximum capacity as 5. When the user configures the AS group, how many instances will Auto Scaling launch?

- A. 3
- B. 0
- C. 5
- D. 2

19. Which of the following are characteristics of Amazon VPC subnets?

- A. Each subnet spans at least 2 Availability Zones to provide a high-availability environment.
- B. Each subnet maps to a single Availability Zone.
- C. CIDR block mask of /25 is the smallest range supported.
- D. By default, all subnets can route between each other, whether they are private or public.
- E. Instances in a private subnet can communicate with the Internet only if they have an Elastic IP.

20. An infrastructure is being hosted in AWS using the following resources
- a) A couple of EC2 Instances serving a Web-Based application
  - b) An Elastic Balancer in front of the EC2 Instances
  - c) An AWS(Amazon Web Service) RDS which has Multi-AZ enabled

Which of the following can be added to the setup to ensure scalability?

Options are :

- A. Add another ELB to the setup.
- B. Add more EC2 Instances to the setup.
- C. Enable Read Replicas for the AWS(Amazon Web Service) RDS.
- D. Add an Auto Scaling Group to the setup.

Answer :Add an Auto Scaling Group to the setup.

21. Which of the following are characteristics of the Auto Scaling service on AWS?  
(Choose 3 answers)

- A. Sends traffic to healthy instances
- B. Responds to changing conditions by adding or terminating Amazon Elastic Compute Cloud (Amazon EC2) instances
- C. Collects and tracks metrics and sets alarms
- D. Delivers push notifications
- E. Launches instances from a specified Amazon Machine Image (AMI)
- F. Enforces a minimum number of running Amazon EC2 instances

*B, E, F. Auto Scaling responds to changing conditions by adding or terminating instances, launches instances from an AMI specified in the launch configuration associated with the Auto Scaling group, and enforces a minimum number of instances in the min-size parameter of the Auto Scaling group.*

22. You create a new subnet and then add a route to your route table that routes traffic out from that subnet to the Internet using an IGW. What type of subnet have you created?

- A. An internal subnet
  - B. A private subnet
  - C. An external subnet
  - D. A public subnet
- D. By creating a route out to the Internet using an IGW, you have made this subnet public.

23. Which of the following can be used to address an Amazon Elastic Compute Cloud (Amazon EC2) instance over the web? (Choose 2 answers)

- A. Windows machine name
- B. Public DNS name
- C. Amazon EC2 instance ID
- D. Elastic IP address

B, D. Neither the Windows machine name nor the Amazon EC2 instance ID can be resolved into an IP address to access the instance.

24. Which of the following functions are that Amazon Route 53 **does not** perform?

- A. DNS Service
- B. Load Balancing
- C. Data Storage
- D. Domain Registration
- E. Health Checking.

*You can use Route 53 to accomplish the three main functions in any combination:*

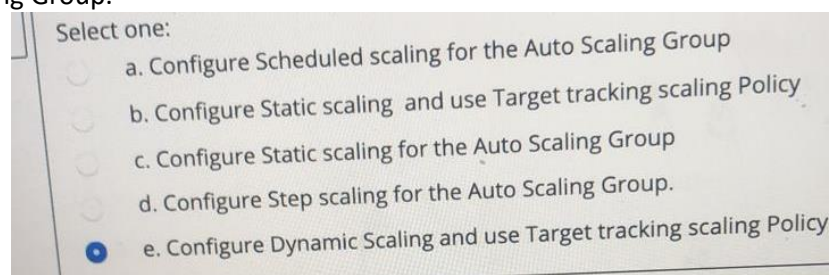
- *Domain Registration*
- *DNS Routing*
- *Health Checking.*

25. An organization is managing a Redshift Cluster in AWS. They need to monitor the performance of the Redshift to ensure that it is performing as efficiently as possible. Which of the following service can be used for achieving this requirement?

- A. Cloudtrail
- B. VPC Flow Logs
- C. Cloudwatch
- D. AWS Trusted Advisor

26. An application is hosted on EC2 Instances. There is a promotional campaign due to start in two weeks for the application. There is a mandate from the management to ensure that no performance problems are encountered due to traffic growth during this time. What action must be taken on the Auto Scaling Group to ensure this requirement can be fulfilled?

- A. Configure step scaling for the Auto Scaling Group.
- B. Configure Dynamic scaling for the Auto Scaling Group
- C. Configure Scheduled scaling for the Auto Scaling Group.
- D. Configure Static scaling for the Auto Scaling Group.



27. You have an application hosted on AWS consisting of EC2 instances launched via an Auto Scaling Group. You notice that the EC2 instances are not scaling up on demand. What checks can be done to ensure that the scaling occurs as expected?

- A. Ensure that the right metrics are being used to trigger the scale out.
- B. Ensure that ELB health checks are being used.
- C. Ensure that the instances are placed across multiple Availability Zones.
- D. Ensure that the instances are placed across multiple regions.

*If your scaling events are not based on the right metrics and do not have the right threshold defined, then the scaling will not occur as you want it to happen.*

*For more information on Auto Scaling Dynamic Scaling, please visit the following URL:*

<https://docs.aws.amazon.com/autoscaling/ec2/userguide/as-scale-based-on-demand.html>

28. You create an Auto Scaling Group which is used to spin up instances On Demand. As an architect, you need to ensure that the instances are pre-installed with a software when they are launched. What are the ways in which you can achieve this? Choose 2 answers from the options given below.

- A. Add the software installation to the configuration for the Auto Scaling Group.
- B. Add the scripts for the installation in the User data section.
- C. Create a golden image and then create a launch configuration.
- D. Ask the IT operations team to install the software as soon as the instance is launched

29. An organization has setup Auto Scaling with ELB. Due to some manual error, one of the instances got rebooted. Thus, it failed the Auto Scaling health check. Auto Scaling has marked it for replacement. How can the system admin ensure that the instance does not get terminated?

- A. Update the Auto Scaling group to ignore the instance reboot event
- B. It is not possible to change the status once it is marked for replacement
- C. Manually add that instance to the Auto Scaling group after reboot to avoid replacement
- D. Change the health of the instance to healthy using the Auto Scaling commands

*After an instance has been marked unhealthy by Auto Scaling, as a result of an Amazon EC2 or ELB health check, it is almost immediately scheduled for replacement as it will never automatically recover its health. If the user knows that the instance is healthy then he can manually call the SetInstanceHealth action (or the as-set instance- health command from CLI. to set the instance's health status back to healthy. Auto Scaling will throw an error if the instance is already terminating or else it will mark it healthy.*

30. A user has created a subnet with VPC and launched an EC2 instance in that subnet with only default settings. Which of the below mentioned options is ready to use on the EC2 instance as soon as it is launched?

- A. Elastic IP    B. Private IP    C. Public IP    D. Internet gateway

*Explanation:*

*A Virtual Private Cloud (VPC. is a virtual network dedicated to a user-s AWS account. A subnet is a range of IP addresses in the VPC. The user can launch the AWS resources into a subnet. There are two supported platforms into which a user can launch instances: EC2-Classic and EC2-VPC. When the user launches an instance which is not a part of the non-default subnet, it will only have a private IP assigned to it. The instances part of a subnet can communicate with each other but cannot communicate over the internet or to the AWS services, such as RDS / S3.*

31. An instance is launched into a VPC subnet with the network ACL configured to allow all inbound traffic and deny all outbound traffic. The instance's security group is configured to allow SSH from any IP address and deny all outbound traffic. What changes need to be made to allow SSH access to the instance?

- A. The outbound security group needs to be modified to allow outbound traffic.  
B. The outbound network ACL needs to be modified to allow outbound traffic.  
C. Nothing, it can be accessed from any IP address using SSH.  
D. Both the outbound security group and outbound network ACL need to be modified to allow outbound traffic.

32. A user has setup an Auto Scaling group. The group has failed to launch a single instance for more than 24 hours. What will happen to Auto Scaling in this condition?

- A. Auto Scaling will keep trying to launch the instance for 72 hours
- B. Auto Scaling will suspend the scaling process
- C. Auto Scaling will start an instance in a separate region
- D. The Auto Scaling group will be terminated automatically

*Explanation:*

*If Auto Scaling is trying to launch an instance and if the launching of the instance fails continuously, it will suspend the processes for the Auto Scaling groups since it repeatedly failed to launch an instance. This is known as an administrative suspension. It commonly applies to the Auto Scaling group that has no running instances which is trying to launch instances for more than 24 hours, and has not succeeded in that to do so.*

33. An application consists of the following architecture:

- a. EC2 Instances in a single AZ behind an ELB
- b. A NAT Instance which is used to ensure that instances can download updates from the Internet

Which of the following can be used to ensure better fault tolerance in this setup?  
Choose 2 answers from the options given below.

- A. Add more instances in the existing Availability Zone.
- B. Add an Auto Scaling Group to the setup.
- C. Add more instances in another Availability Zone.
- D. Add another ELB for more fault tolerance.

34. Your architecture for an application currently consists of EC2 Instances sitting behind a classic ELB. The EC2 Instances are used to serve an application and are accessible through the internet. What can be done to improve this architecture in the event that the number of users accessing the application increases?

- A. Add another ELB to the architecture.
- B. Use Auto Scaling Groups.
- C. Use an Application Load Balancer instead.
- D. Use the Elastic Container Service.

*AWS Documentation mentions the following: AWS Auto Scaling monitors your applications and automatically adjusts capacity to maintain steady, predictable performance at the lowest possible cost.*

*Using AWS Auto Scaling, it is easy to setup application scaling for multiple resources across multiple services in minutes.*

35. There is an urgent requirement to monitor some database metrics for a database hosted on AWS and send notifications. Which AWS services can accomplish this?

- A. Amazon Simple Email Service
- B. Amazon CloudWatch
- C. Amazon Simple Queue Service
- D. Amazon Simple Notification Service

*Amazon CloudWatch will be used to monitor the IOPS metrics from the RDS Instance and Amazon Simple Notification Service will be used to send the notification if any alarm is triggered.*

36. Which of the following statements are true with respect to VPC?

- A. A subnet can have multiple route tables associated with it.
- B. A network ACL can be associated with multiple subnets.
- C. A route with target "local" on the route table can be edited to restrict traffic within VPC.
- D. Subnet's IP CIDR block can be same as the VPC CIDR block.

37. To scale up the AWS resources using manual Auto Scaling, which of the below mentioned parameters should the user change?

- A. Maximum capacity
- B. Desired capacity
- C. Preferred capacity
- D. Current capacity

*Explanation:*

*The Manual Scaling as part of Auto Scaling allows the user to change the capacity of Auto Scaling group. The user can add / remove EC2 instances on the fly. To execute manual scaling, the user should modify the desired capacity.*

*AutoScaling will adjust instances as per the requirements. If the user is trying to CLI, he can use command:*

*as-set-desired-capacity <Auto Scaling Group Name> --desired-capacity <New Capacity>*

<http://docs.aws.amazon.com/AutoScaling/latest/DeveloperGuide/as-manual-scaling.html>

38. Elastic Load Balancing supports which of the following types of load balancers?

- A. Cross-region
- B. Internet-facing
- C. Interim
- D. Itinerant
- E. Internal
- F. Hypertext Transfer Protocol Secure (HTTPS) using Secure Sockets Layer (SSL)

*B, E, F. Elastic Load Balancing supports Internet-facing, internal, and HTTPS load balancers.*

39. A company has an application hosted in AWS. This application consists of EC2 Instances which sit behind an ELB. The following are the requirements from an administrative perspective:

- a) Ensure notifications are sent when the read requests go beyond 1000 requests per minute
- b) Ensure notifications are sent when the latency goes beyond 10 seconds
- c) Monitor all API activities on the AWS resources Which of the followings can be used to satisfy these requirements? (SELECT TWO)

- A. Use CloudTrail to monitor the API Activity
- B. Use CloudWatch logs to monitor the API Activity
- C. Use CloudWatch metrics for the metrics that need to be monitored as per the requirement and set up an alarm activity to send out notifications when the metric reaches the set threshold limit
- D. Use custom log software to monitor the latency and read request to the ELB

40. When Auto Scaling is launching a new instance based on condition, which of the below mentioned policies will it follow?

- A. Based on the criteria defined with cross zone Load balancing
- B. Launch an instance which has the highest load distribution
- C. Launch an instance in the AZ with the fewest instances
- D. Launch an instance in the AZ which has the highest instances

*Explanation:*

*Auto Scaling attempts to distribute instances evenly between the Availability Zones that are enabled for the user's Auto Scaling group. Auto Scaling does this by attempting to launch new instances in the Availability Zone with the fewest instances.*

[http://docs.aws.amazon.com/AutoScaling/latest/DeveloperGuide/AS\\_Concepts.html](http://docs.aws.amazon.com/AutoScaling/latest/DeveloperGuide/AS_Concepts.html)

41. You have decided to change the instance type for instances running in your application tier that are using Auto Scaling. In which area below would you change the instance type define?

- A. Auto Scaling policy
- B. Auto Scaling group
- C. Auto Scaling tags
- D. Auto Scaling launch configuration

*Explanation:*

*When selecting an instance to terminate when a scaling condition is met, Auto Scaling attempts to preserve instances with the current launch configuration, and will thus terminate instances that do not have the current launch configuration. When more than one instance meets this criterion, Auto Scaling will terminate the instance running for the longest portion of a billable instance-hour (without running over). Optionally, you can configure a policy to terminate the oldest or newest instance instead. To target a specific instance for immediate termination, you can also use the `TerminateInstanceInAutoScalingGroup` API.*

42. A user has launched an EC2 instance. The user is planning to setup the CloudWatch alarm. Which of the below mentioned actions is not supported by the CloudWatch alarm?

- A. Notify the Auto Scaling launch config to scale up
- B. Send an SMS using SNS
- C. Notify the Auto Scaling group to scale down
- D. Stop the EC2 instance

*Explanation:*

*A user can create a CloudWatch alarm that takes various actions when the alarm changes state. An alarm watches a single metric over the time period that the user has specified, and performs one or more actions based on the value of the metric relative to a given threshold over a number of time periods. The actions could be sending a notification to an Amazon Simple Notification Service topic (SMS, Email, and HTTP end point.), notifying the Auto Scaling policy or changing the state of the instance to Stop/Terminate.*

43. An organization is measuring the latency of an application every minute and storing data inside a file in the JSON format. The organization wants to send all latency data to AWS CloudWatch. How can the organization achieve this?

- A. The user has to parse the file before uploading data to CloudWatch
- B. It is not possible to upload the custom data to CloudWatch
- C. The user can supply the file as an input to the CloudWatch command
- D. The user can use the CloudWatch Import command to import data from the file to CloudWatch

*AWS CloudWatch supports the custom metrics. The user can always capture the custom data and upload the data to CloudWatch using CLI or APIs. The user always has to include the namespace as part of the request. If the user wants to upload the custom data from a file, he can supply file name along with the parameter --metric-data to command put-metric-data.*

44. Which of the following are the minimum required elements to create an Auto Scaling launch configuration?

- A. Launch configuration name
- B. Amazon Machine Image (AMI)
- C. Instance type
- D. Security Group
- E. Key Pair

Only the launch configuration name, AMI, and instance type are needed to create an Auto Scaling launch configuration. Identifying a key pair, security group, and a block device mapping are optional elements for an Auto Scaling launch configuration.

45. As the cloud administrator of your company, you notice that one of EC2 instances is restarting frequently. There is a need to troubleshoot and analyse the system logs. What can be used in AWS to store and analyze the log files from the EC2 Instance? Choose one answer from the options below.

- A. AWS SQS
- B. AWS S3
- C. AWS CloudTrail
- D. AWS CloudWatch Logs

*You can use Amazon CloudWatch Logs to monitor, store, and access your log files from Amazon Elastic Compute Cloud (Amazon EC2) instances, AWS CloudTrail, and other sources.*

46. While reviewing the Auto Scaling events for your application, you notice that your application is scaling up and down multiple times in the same hour. What design choice could you make to optimize costs while preserving elasticity?

- A. Modify the Auto Scaling policy to use scheduled scaling actions
- B. Modify the Auto Scaling group termination policy to terminate the oldest instance first.
- C. Modify the Auto Scaling group cool-down timers.
- D. Modify the Amazon CloudWatch alarm period that triggers your Auto Scaling scale down policy.
- E. Modify the Auto Scaling group termination policy to terminate the newest instance first.

47. For custom CloudWatch metrics, what is the minimum granularity in terms of time that CloudWatch can monitor

- A. 5 minutes
- B. 3 minutes
- C. 2 minutes
- D. 1 minute

48. By default, EC2 monitoring carried out by CloudWatch monitors which metrics?

- A. Memory
- B. CPU
- C. Status
- D. Disk
- E. Network

49. A company is storing data on Amazon Simple Storage Service (S3). The company's security policy mandates that data is encrypted at rest. Which of the following methods can achieve this?

- A. Use Amazon S3 server-side encryption with AWS Key Management Service managed keys.
- B. Use Amazon S3 server-side encryption with customer-provided keys.
- C. Use Amazon S3 server-side encryption with EC2 key pair.
- D. Use Amazon S3 bucket policies to restrict access to the data at rest.
- E. Encrypt the data on the client-side before ingesting to Amazon S3 using their own master key.
- F. Use SSL to encrypt the data while in transit to Amazon S3.

50. A user is planning to use AWS CloudFormation for his automatic deployment requirements. Which of the below mentioned components are required as a part of the template?

- A. Parameters
- B. Outputs
- C. Template version
- D. Resources

*Explanation:*

*AWS CloudFormation is an application management tool which provides application modelling, deployment, configuration, management and related activities. The template is a JSON-format, text-based file that describes all the AWS resources required to deploy and run an application. It can have*

option fields, such as Template Parameters, Output, Data tables, and Template file format version. The only mandatory value is Resource. The user can define the AWS services which will be used/ created by this template inside the Resource section

51. Which of the following can be used to trigger scaling up or down for an Auto Scaling group

- A. CloudWatch      B. The AWS console      C. SNS      D. S3      E. Route 53

The Auto Scaling group in your Elastic Beanstalk environment uses two Amazon CloudWatch alarms to trigger scaling operations. The default triggers scale when the average outbound network traffic from each instance is higher than 6 MB or lower than 2 MB over a period of five minutes. To use Amazon EC2 Auto Scaling effectively, configure triggers that are appropriate for your application, instance type, and service requirements. You can scale based on several statistics including latency, disk I/O, CPU utilization, and request count.

52. An auto scaling group may use :

- A. On-Demand Instances  
B. Stopped instances  
C. Spot Instances  
D. On-premises instances  
E. Already running instances if they use the same Amazon Machine Image (AMI) as the Auto Scaling group's launch configuration and are not already part of another Auto Scaling group

A, C. An Auto Scaling group may use On-Demand and Spot Instances. An Auto Scaling group may not use already stopped instances, instances running someplace other than AWS, and already running instances not started by the Auto Scaling group itself.

53. What does Amazon CloudFormation provide?

- A. The ability to setup Autoscaling for Amazon EC2 instances.  
B. None of these.  
C. A template resource creation for Amazon Web Services.  
D. A template to map network resources for Amazon Web Services.

54. Your company currently has a set of EC2 instances hosted in AWS. The states of these instances need to be monitored ~~and each state change needs to be recorded~~. Which of the following can help fulfill this requirement?

- A. Use CloudWatch logs to store the state change of the instances

- B. Create an amazon CloudWatch alarm that monitors and Amazon EC2 instance / **Use CloudWatch Events to monitor the state change of events**
- C. Use SQS to trigger a record to be added to a DynamoDB table
- D. Use AWS Lambda to store a change record in DynamoDB table

Correct Answer is ~~a and b~~ ~~Use CloudWatch logs to store the state change of the instances AND~~  
 Create an amazon CloudWatch alarm that monitors and Amazon EC2 instance or **Use CloudWatch Events to monitor the state change of events**

**Events** – An event indicates a change in your AWS environment. AWS resources can generate events when their state changes. For example, Amazon EC2 generates an event when the state of an EC2 instance changes from pending to running, and Amazon EC2 Auto Scaling generates events when it launches or terminates instances. AWS CloudTrail publishes events when you make API calls. You can generate custom application-level events and publish them to CloudWatch Events. You can also set up scheduled events that are generated on a periodic basis. For a list of services that generate events, and sample events from each service, see [CloudWatch Events Event Examples From Supported Services](#).

55. A customer is using AWS for Dev and Test. The customer wants to setup the Dev environment with Cloudformation. Which of the below mentioned steps are not required while using Cloudformation?
- A. Create a stack
  - B. **Configure a service**
  - C. Create and upload the template
  - D. Provide the parameters configured as part of the template

*Explanation:*

AWS Cloudformation is an application management tool which provides application modelling, deployment, configuration, management and related activities. AWS CloudFormation introduces two concepts: the template and the stack. The template is a JSON-format, text-based file that describes all the AWS resources required to deploy and run an application. The stack is a collection of AWS resources which are created and managed as a single unit when AWS CloudFormation instantiates a template. While creating a stack, the user uploads the template and provides the data for the parameters if required.

56. Why is the launch configuration referenced by the Auto Scaling group instead of being part of the Auto Scaling group? Select One

- A. It allows you to change the Amazon Elastic Compute Cloud (Amazon EC2) instance type and Amazon Machine Image (AMI) without disrupting the Auto Scaling group.
- B. It facilitates rolling out a patch to an existing set of instances managed by an Auto Scaling group.
- C. It allows you to change security groups associated with the instances launched without having to make changes to the Auto Scaling group.

D. All of the above

E. None of the above

*D - A, B, and C are all true statements about launch configurations being loosely coupled and referenced by the Auto Scaling group instead of being part of the Auto Scaling group.*

57. Because of the extensibility limitations of striped storage attached to Windows Server, Amazon RDS does not currently support increasing storage on a \_\_\_\_\_ DB Instance.

A. SQL Server

B. MySQL

C. Oracle

D. Oracle & MySQL

E. SQL Server & MySQL

58. An organization is planning to use AWS for their production roll out. The organization wants to implement automation for deployment such that it will automatically create a LAMP stack, download the latest PHP installable from S3 and setup the ELB. Which of the below mentioned AWS services meets the requirement for making an orderly deployment of the software?

A. AWS Elastic Beanstalk

B. AWS Cloudfront

C. AWS Cloudformation

D. AWS DevOps

AWS Cloudformation is an application management tool which provides application modelling, deployment, configuration, management and related activities.

Cloudformation provides an easy way to create and delete the collection of related AWS resources and provision them in an orderly way. AWS CloudFormation automates and simplifies the task of repeatedly and predictably creating groups of related resources that power the user's applications. AWS Cloudfront is a CDN;

Elastic Beanstalk does quite a few of the required tasks. However, it is a PAAS which uses a ready AMI. AWS Elastic Beanstalk provides an environment to easily develop and run applications in the cloud.

59. What is an isolated database environment running in the cloud (Amazon RDS) called?

- A. DB Storage      B. DB Server      C. DB Unit      **D. DB instance**      E. DB Volume

A **DB instance** is an **isolated database environment running in the cloud**. It is the basic building block of **Amazon RDS**. A **DB instance** can contain multiple user-created **databases**, and can be accessed using the same client tools and applications you might use to access a standalone **database instance**.

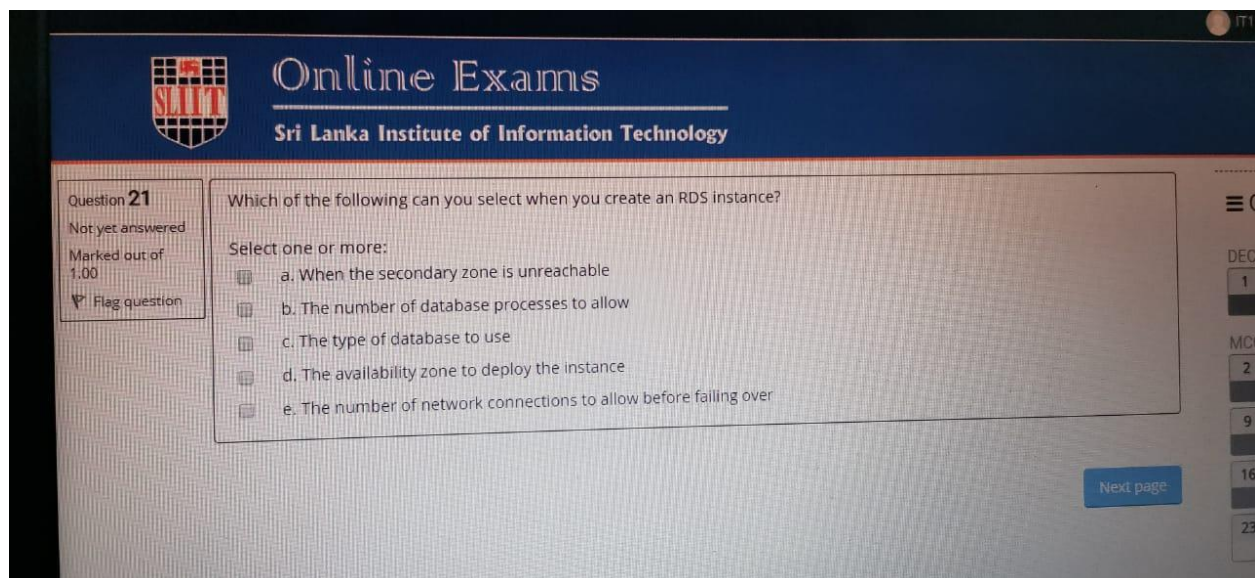
60. You are deploying an application to collect votes for a very popular television show. Millions of users will submit votes using mobile devices. The votes must be collected into a durable, scalable, and highly available data store for real-time public tabulation. Which service should you use?

- A. Amazon DynamoDB**      B. Amazon Redshift      C. Amazon Kinesis

D. Amazon Simple Queue Service

61. Which of the following can you select when you create an RDS instance?

When creating an RDS instance, you can select which availability zone to deploy the instance. Therefore answer D.



62. You have been tasked with ensuring that data stored in your organization's RDS instance exists in a minimum geographically distributed location. Which of the following solutions are valid approaches?

(Choose two.)

- A. Enable RDS in a Multi-AZ configuration.
- B. Enable RDS in a read replica configuration.
- C. Install a storage gateway with stored volumes.
- D. Enable RDS in a cross-region read replica configuration

A,D - Multi-AZ setup is the easiest solution, and the most common. Turning on read replicas (option B) is not a guarantee, as read replicas are not automatically installed in different AZs or regions. However, with option D, a cross-region replica configuration will ensure multiple regions are used. A storage gateway (option C) is backed by S3, not RDS.

63. Which of the following notification endpoints or clients are supported by Amazon Simple Notification Service?

Choose 2 answers

- A. Email
- B. CloudFront distribution
- C. File Transfer Protocol
- D. Short Message Service
- E. Simple Network Management Protocol

Explanation: <http://docs.aws.amazon.com/sns/latest/dg/welcome.html>

64. A company is planning to run a number of Admin related scripts using the AWS Lambda service. There is a need to detect errors that occur while the scripts run. How can this be accomplished in the most effective manner?

- A. Use Cloudwatch metrics and logs to watch for errors
- B. Use Cloudtrail to monitor for errors
- C. Use the AWS Config service to monitor for errors
- D. Use the AWS inspector service to monitor for errors

### Explanation

The AWS Documentation mentions the following

AWS Lambda automatically monitors Lambda functions on your behalf, reporting metrics through Amazon CloudWatch. To help you troubleshoot failures in a function. Lambda logs all requests handled by your function and also automatically stores logs generated by your code through Amazon CloudWatch Logs.

Option B,C and D are all invalid because these services cannot be used to monitor for errors. For more information on Monitoring Lambda functions, please visit the following URL:

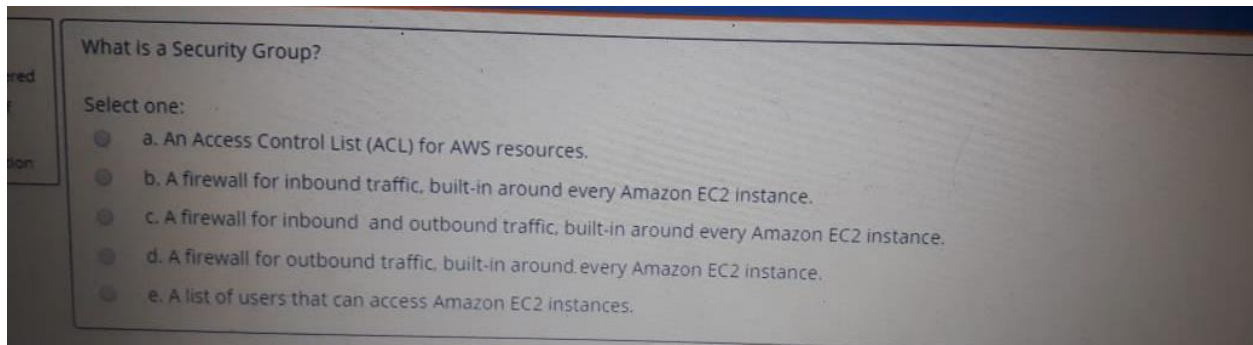
<https://docs.aws.amazon.com/lambda/latest/dg/monitoring-functions-logs.html> The correct answer is: Use Cloudwatch metrics and logs to watch for errors

65. Which of the following **cannot be used** in Amazon EC2 to control who has access to specific Amazon EC2 instances?

- A. Security Groups      B. IAM System      C. SSH keys      D. Windows passwords

66. What is a Security Group?

- A. None of these.  
B. A list of users that can access Amazon EC2 instances.  
C. An Access Control List (ACL) for AWS resources.  
D. A firewall for inbound traffic, built-in around every Amazon EC2 instance.



67. A company has an application hosted in AWS. This application consists of EC2 Instances which sit behind an ELB. The following are the requirements from an administrative perspective:

- a) Ensure notifications are sent when the read requests go beyond 1000 requests per minute
- b) Ensure notifications are sent when the latency goes beyond 10 seconds

Which of the followings can be used to satisfy these requirements? (SELECT ONE)

A. Use CloudTrail to monitor the API Activity.

B. Use CloudWatch logs to monitor the API Activity.

C. Use CloudWatch metrics for the metrics that needs to be monitored as per the requirement and set up an alarm activity to send out notifications when the metric reaches the set threshold limit.

D. Use a custom log software to monitor the latency and read requests to the ELB.

*AWS CloudTrail can be used to monitor the API calls.*

*For more information on CloudTrail, please visit the following URL: <https://aws.amazon.com/cloudtrail/>*

***When you use CloudWatch metrics for an ELB, you can get the amount of read requests and latency out of the box.***

68. Which two AWS services provide out-of-the-box user configuration automatic backup-as-a-service and backup rotation options?

A. Amazon S3

B. Amazon RDS

C. Amazon EBS

D. Amazon Red shift

69. Which is a recommended way to protect Access Keys?

A. Train developers how to better protect their access keys.

B. Define IAM policies.

C. Enable CloudWatch notifications.

D. All of the above.

70. Which of the following will ensure that data on your RDS instance is encrypted?

A. Use client-side encryption keys.

B. Enable encryption on the running RDS instance via the AWS API.

C. Encrypt the instance on which RDS is running.

D. None of these will encrypt all data on the instance.

*D - You cannot encrypt a running RDS instance, so B is incorrect, and you have no access to the underlying instance for RDS, so C is also incorrect. Option A sounds possible, but it will not address any data created by the database itself (such as indices, references to other data in the database, etc.). The only way to encrypt an RDS instance is to encrypt it at creation of the instance.*

71. Which